

RYNE FRANK SHELTON

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Professional Summary

Electrical and Computer Engineer specializing in semiconductor failure analysis, device characterization, and engineering automation. Experienced with nanoprobeing, TEM, SEM, EBAC, Electrical Failure Analysis (EFA), Physical Failure Analysis (PFA), transistor-level electrical characterization, defect localization, and root-cause investigation for advanced-node semiconductor technologies. Proven experience developing Python-based automation and machine-learning solutions across semiconductor manufacturing, research, and engineering environments. Technical leader and engineering mentor currently pursuing an M.S. in Electrical Engineering with a 4.00 GPA.

Core Technical Competencies

Semiconductor Failure Analysis: Nanoprobeing, TEM, SEM, EBAC, Electrical Failure Analysis (EFA), Physical Failure Analysis (PFA), Root-Cause Analysis, Defect Localization, SRAM Characterization, Device Characterization

Electrical & Device Engineering: Transistor-Level Electrical Characterization, Signal-Path Verification, RF Instrumentation, Electrical Measurement, LTspice

Semiconductor Manufacturing & Equipment: Advanced-Node Semiconductor Technologies, Vacuum Systems, Dry-Etch Equipment, RF Systems, Mass Flow Controllers, Tool Calibration, Preventive Maintenance, Corrective Maintenance, Cleanroom Operations

Engineering Automation & Software: Python, C/C++, MicroPython, SQL, Linux, Git, Power BI, Excel, LaTeX

Machine Learning & Data Analytics: TensorFlow, PyTorch, Scikit-Learn, Anomaly Detection, Pattern Recognition, Machine Vision

Professional Experience

Intel Corporation — Hillsboro, OR

May 2025 – Dec 2025

TEM/SEM Analysis Technician

- Investigated advanced-node semiconductor failures using nanoprobeing, TEM, and SEM to support root-cause identification across logic, SRAM, and process-monitor structures.
- Performed transistor-level electrical characterization, signal-path verification, and fault localization on complex semiconductor devices.
- Correlated electrical signatures with physical defects using Electrical Failure Analysis (EFA) and Physical Failure Analysis (PFA) methodologies to support yield and product-engineering investigations.
- Conducted TEM analysis and measurement extraction from FIB-prepared samples to identify structural and material failure mechanisms.
- Delivered technical findings reports supporting engineering root-cause analysis and corrective-action decisions.
- Maintained high-throughput analytical operations while processing 5–10 semiconductor samples daily.

RYSYS — Portland, OR

Jan 2025 – Present

Founder & Lead Developer

- Built cloud-based asset-management platforms supporting traceability, automation, and operational visibility.
- Automated verification workflows to reduce manual effort and strengthen software governance.

Hitachi High-Tech — Hillsboro, OR

Mar 2024 – Jul 2024

Process Technician II – Software & Automation Focus

- Developed a Python-based FOUP scheduling system to automate operations within a 24/7 semiconductor laboratory.
- Reduced equipment setup time by more than 80% through workflow automation.
- Improved tool readiness and process stability through calibration of etch, RF, vacuum, and gas-delivery systems.

Portland State University & U.S. Geological Survey — Portland, OR

Jan 2021 – Aug 2023

Engineering Learning Center Manager & Machine Learning Consultant

- Developed machine-learning models for anomaly detection and pattern recognition in large geophysical datasets.
- Led engineering learning-center operations while supervising 10–15 student employees.
- Mentored engineering students in electrical and computer engineering coursework and laboratory techniques.

Datalogic — Eugene, OR

Mar 2022 – Aug 2022

R&D Engineering Intern

- Developed CNN and LSTM machine-vision models for Data Matrix localization and optical inspection applications.

- Managed HVAC, refrigeration, instrumentation, preventive maintenance, and contractor coordination activities.

Education

Portland State University — Portland, OR <i>Master of Science, Electrical Engineering (Analog/RF)</i>	2024 – Present
Portland State University — Portland, OR <i>Bachelor of Science, Electrical & Computer Engineering</i> GPA: 4.00/4.00	2021 – 2024
Portland Community College — Portland, OR <i>Associate of Science, Electrical Engineering</i>	2019 – 2021

Leadership, Research & Honors

- NASA Oregon Space Grant Consortium (OSGC/SCORE) Principal Investigator
- Engineering Learning Center Manager supervising 10–15 student employees
- Multiple Engineering Co-op Program (MECOP) Participant
- Computer Action Team (CAT) Member
- Louis Stokes Alliance for Minority Participation (LSAMP) Scholar
- Student Senator and Student Fee Committee Member
- Mathematics, Engineering and Science Achievement (MESA) College Mentor